

Ship Your Code

*while
it's*

Hot



**Albert
Zak**

1. **Hot Swap**
2. **Release**
3. **Upgrade**
4. **Deploy**
5. **Pitfalls**
6. **Automate**



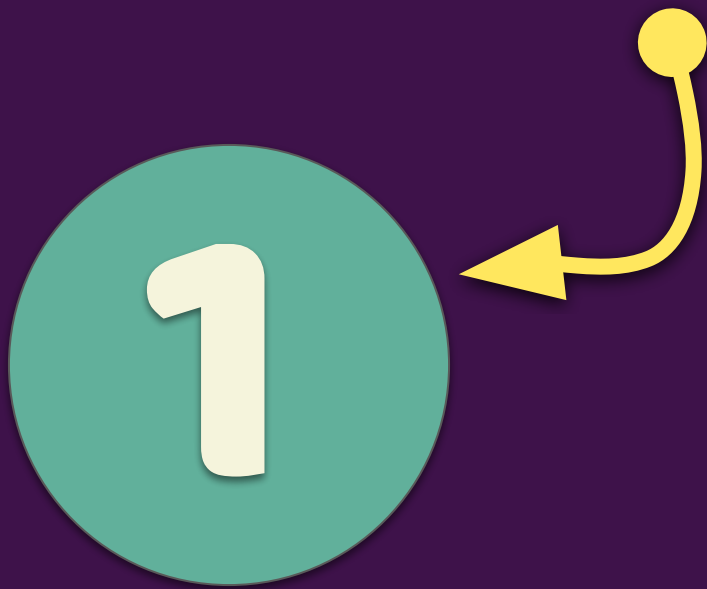
 **mix start**

Hot
swap





current



old



current



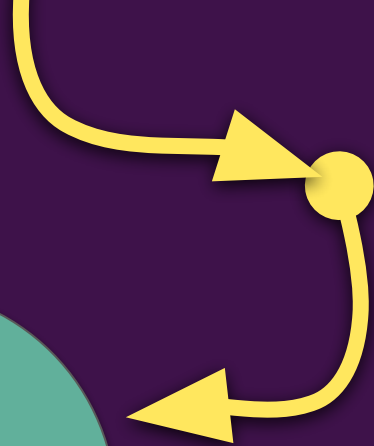
old



current



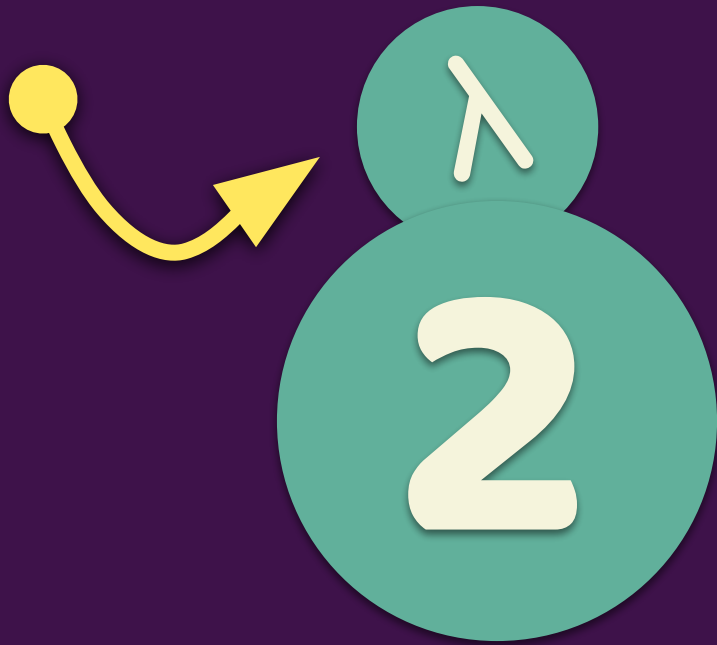
old



current



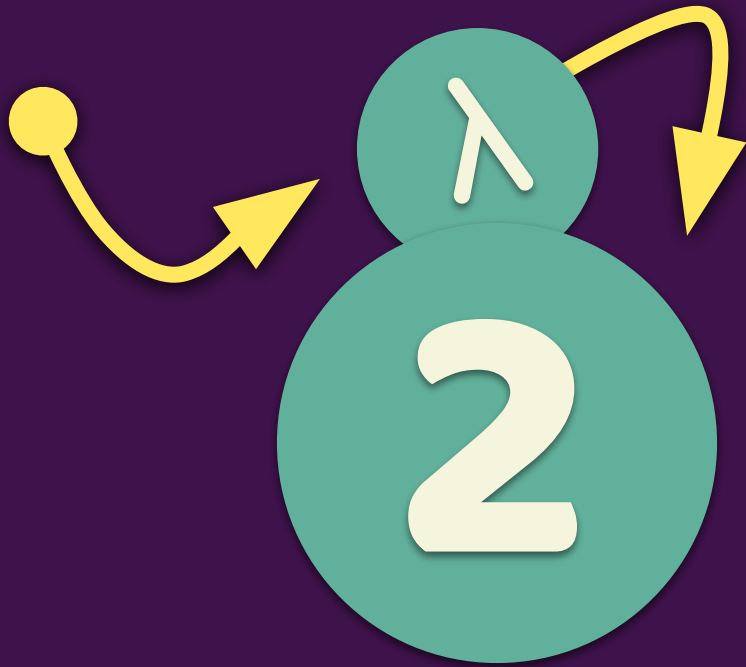
old



current



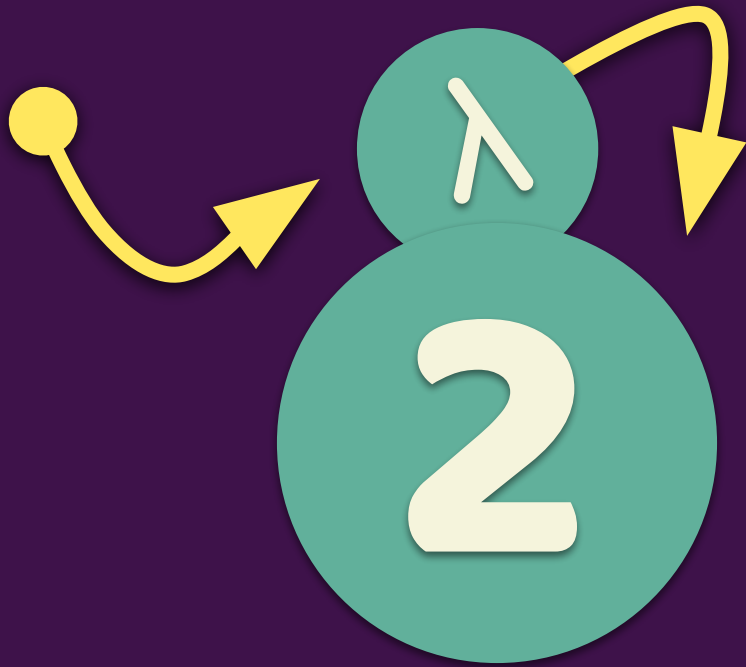
old



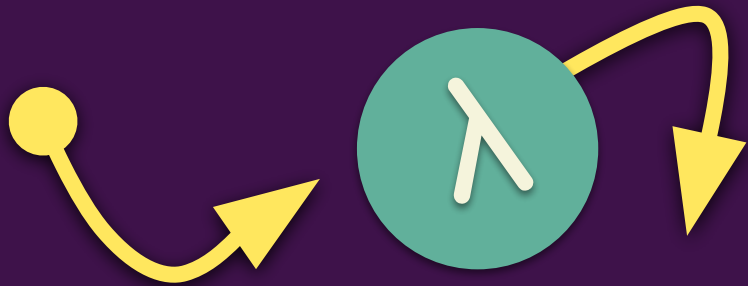
current



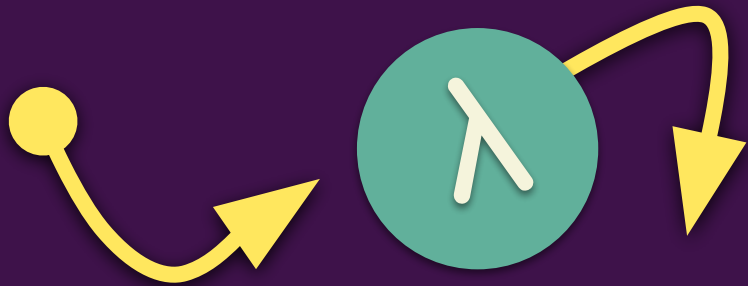
old



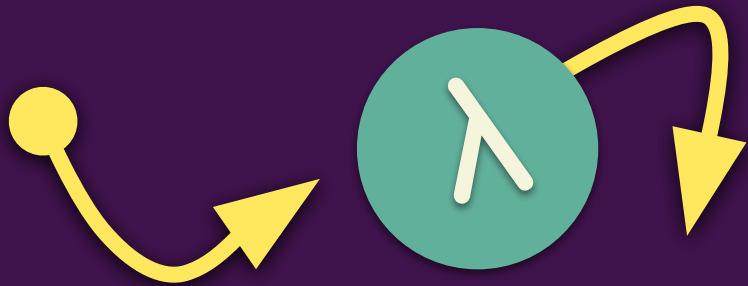
current



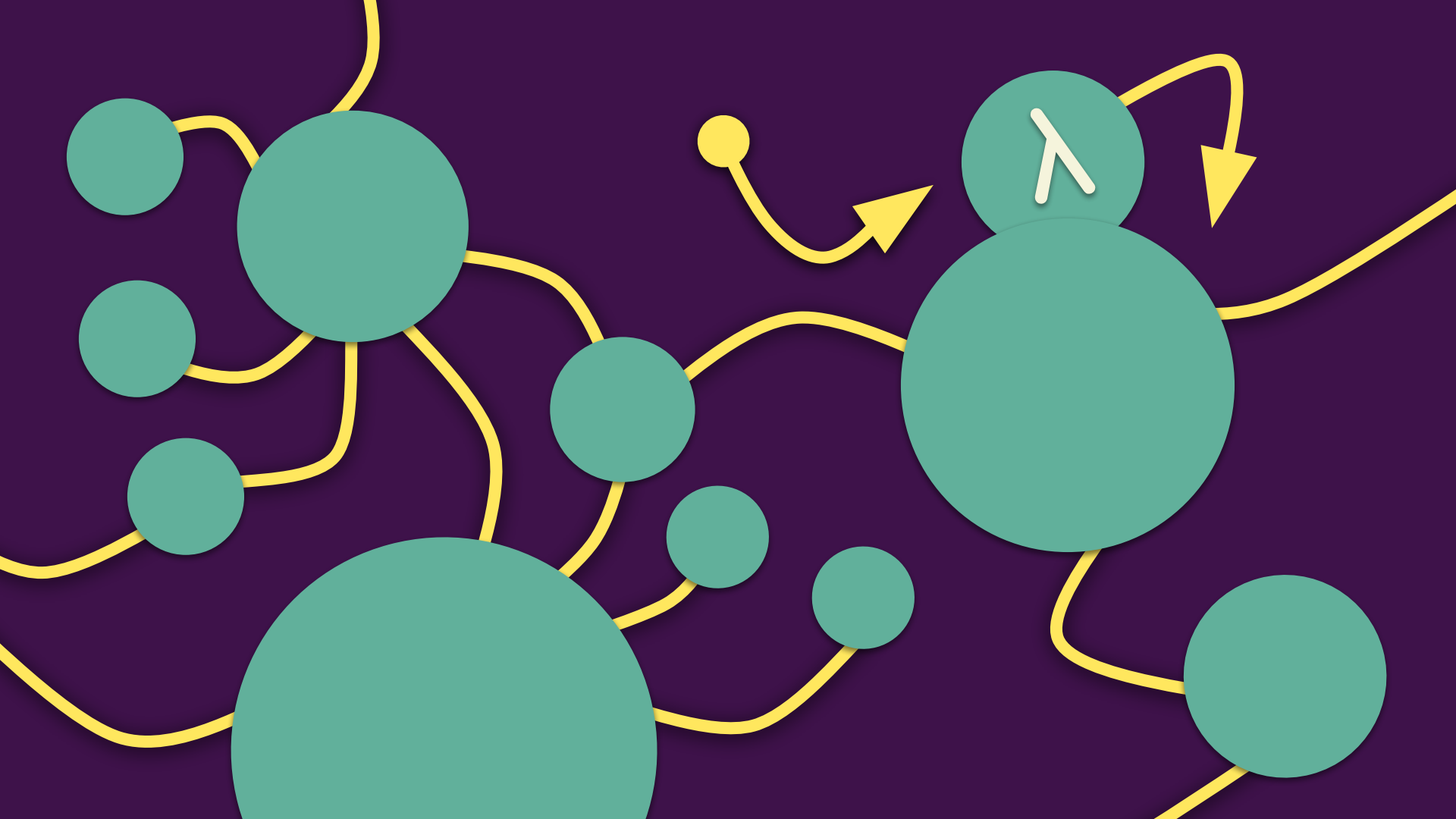
```
code_change(old_vsn,  
            old_state,  
            extra)
```



```
code_change({:down, v},  
            old_state,  
            extra)
```



`{:ok, new_state}`
`{:error, reason}`





Release

bin/

erts-9.3/

lib/

releases/

releases/

hey.rel

RELEASES

0.0.1/

releases/

hey.rel

RELEASES

0.0.2/

relup

:systools

:release_handler

upgrade



***.appup**
***.appup**
***.appup**



relup

```
{ "0.0.2",  
  [ { "0.0.1",  
      [ { load_module, 'Elixir.Hey', [] } ] },  
    [ { "0.0.1",  
        [ { load_module, 'Elixir.Hey', [] } ] } ] } ] }.
```



```
{ "0.0.2",  
  [ { "0.0.1", [],  
    [ { load_object_code,  
      { hey, "0.0.2", ['...'] } } },  
    point_of_no_return,  
    { load, { 'Elixir.Hey',  
      brutal_purge, brutal_purge } },  
    { load, { ... } }  
  ],  
  ...  
}
```

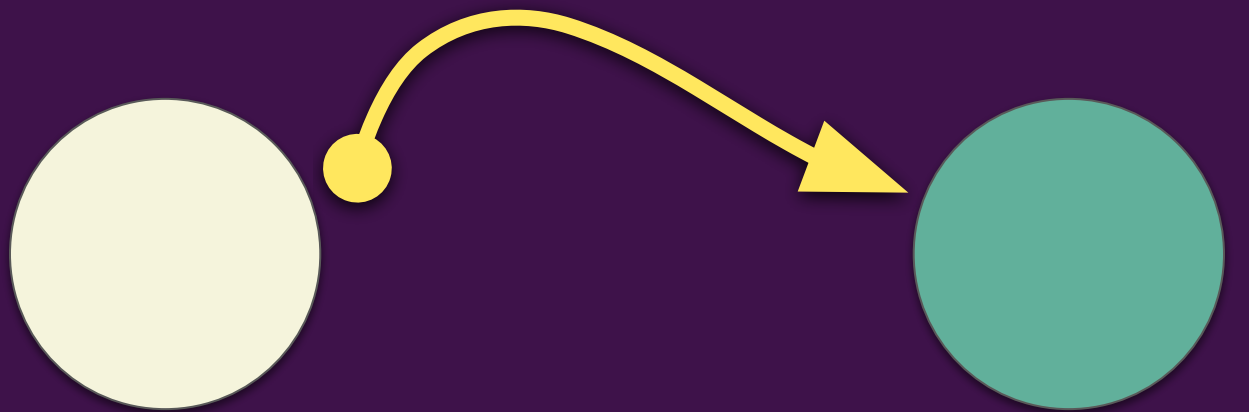
apply

restart_new_emulator

sync_nodes

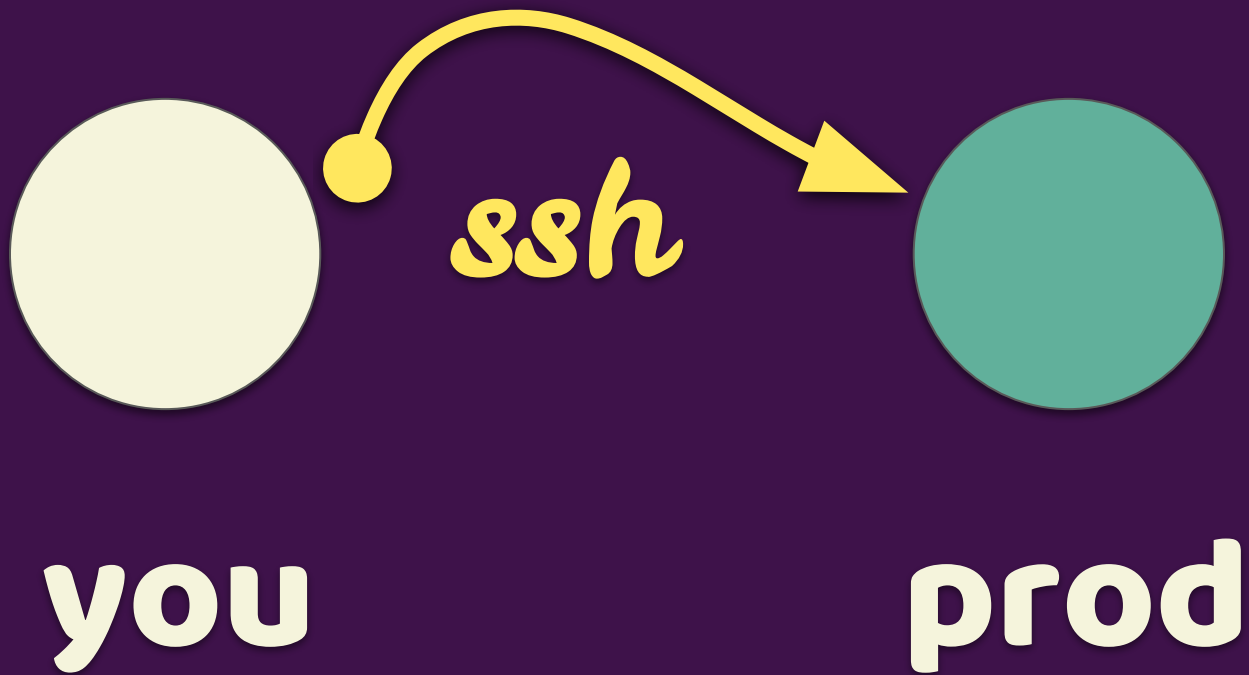
Deploy ↗

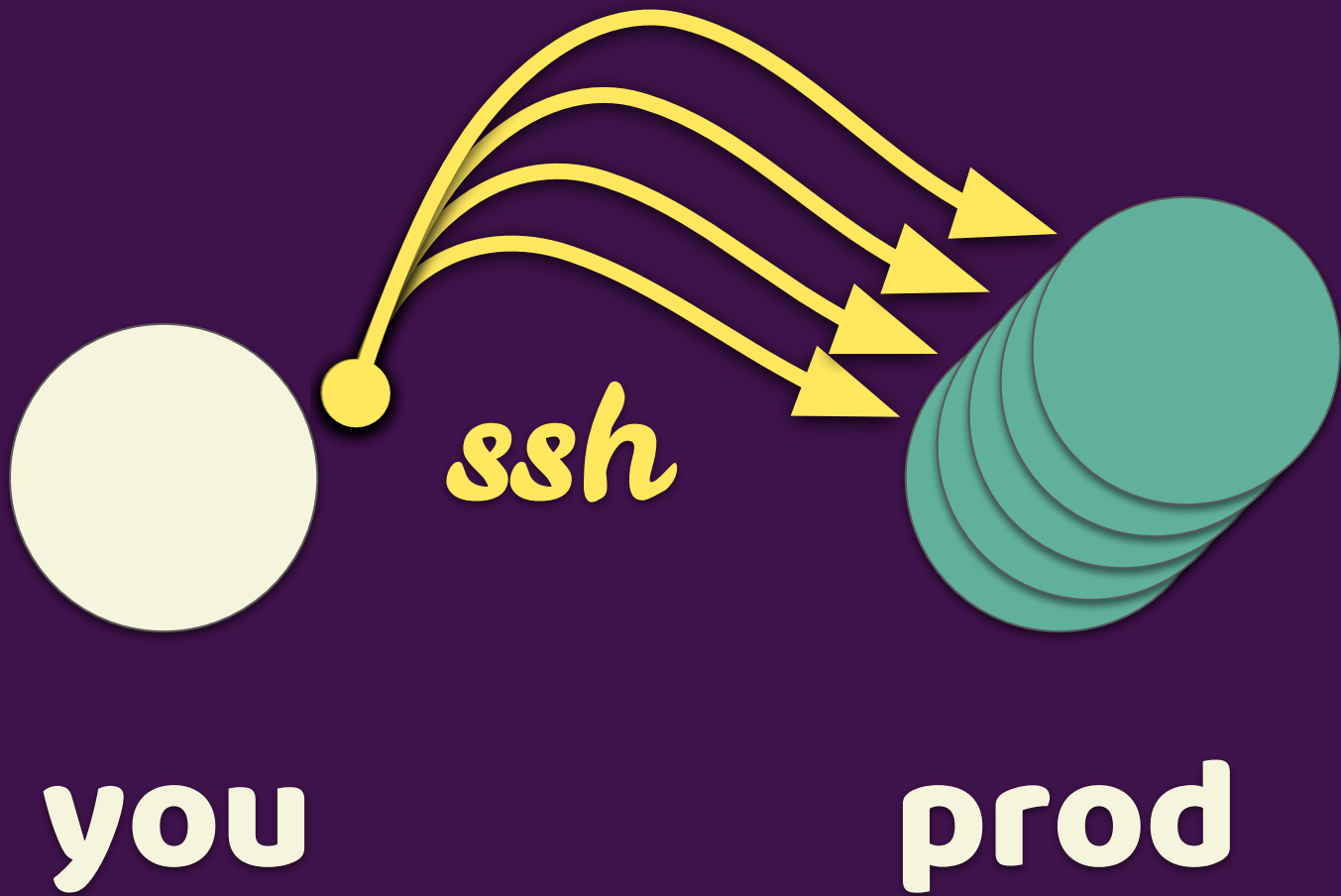


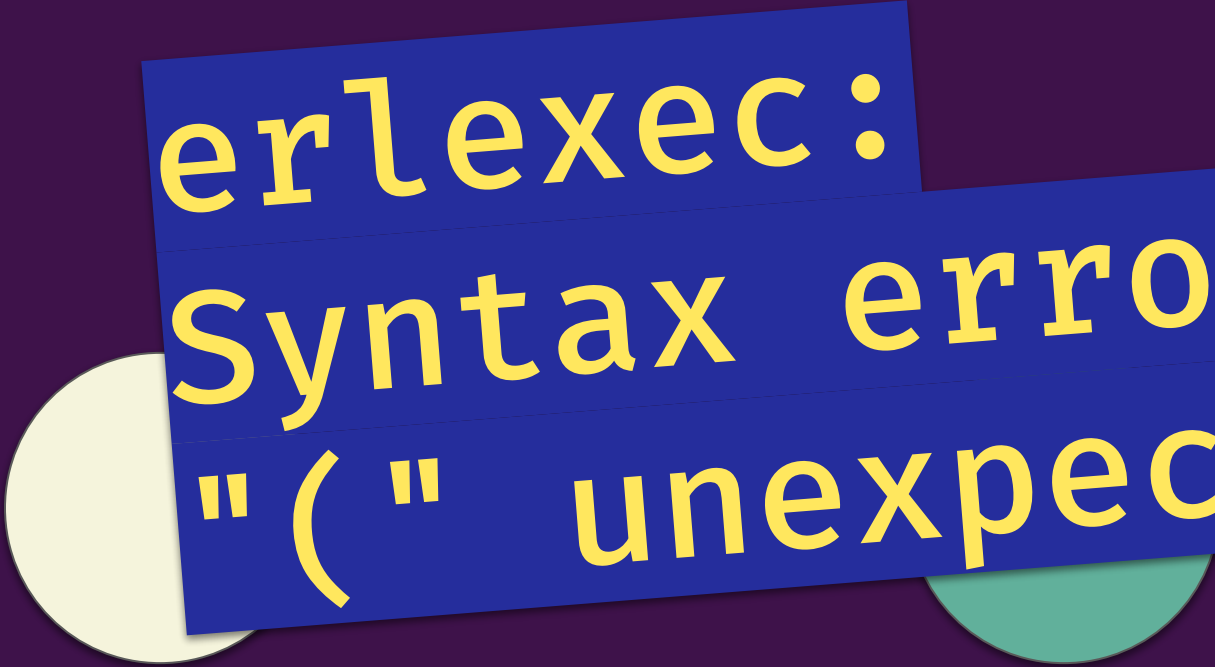


you

prod







erlexec:

Syntax error:

"(" unexpected

you

prod

Containers



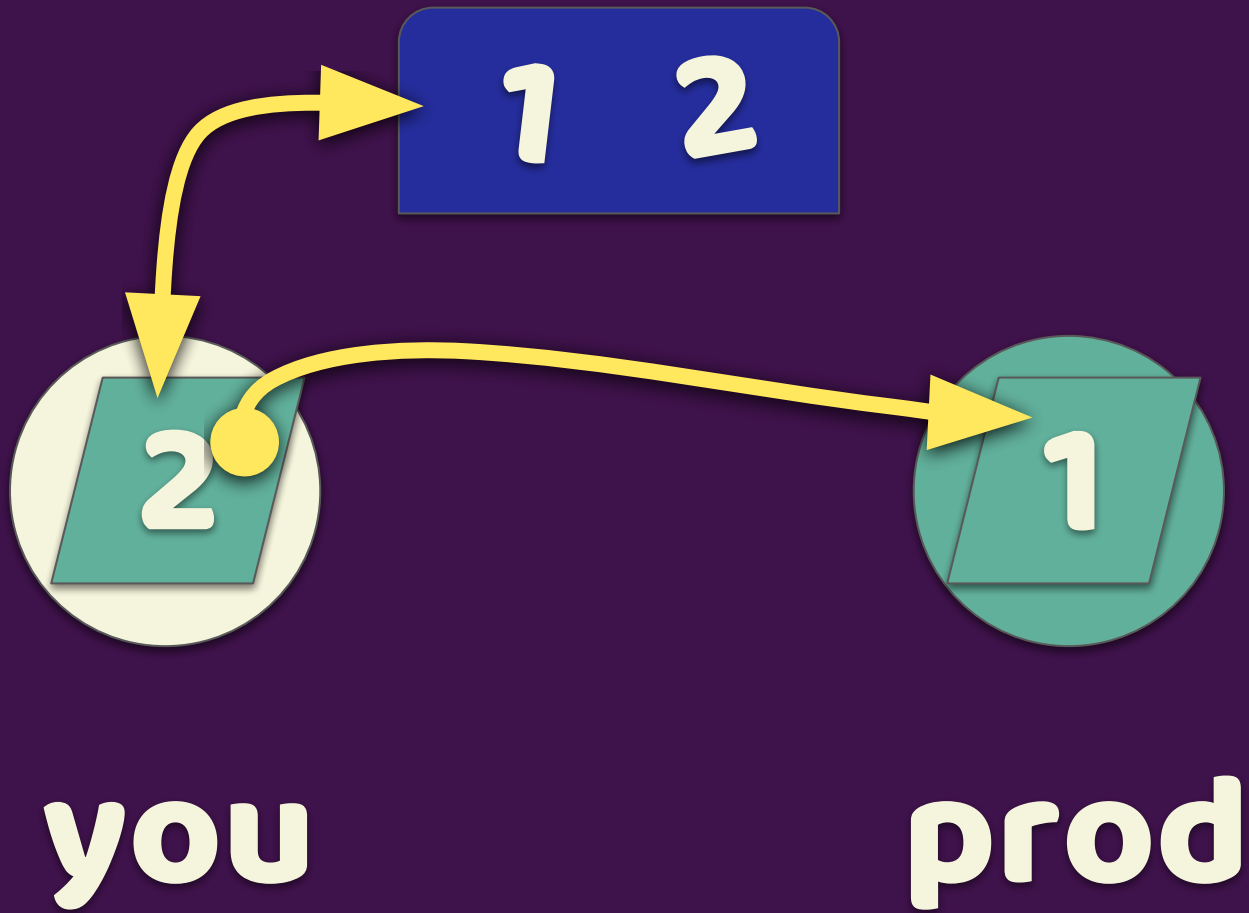
you

prod



you

prod



Pitfalls



```
receive do
```

```
  :old_msg → :ok
```

```
after
```

```
  1000 → Mod.loop()
```

```
end
```

receive do

 :old_msg → :ok

after

 1000 → Mod.loop()

end

fn

$\{m, f, a\}$



Mix Tasks



Migrations

`config.exs` `{:system, v}`

Config `init/2` `ENV`
`Vault?`

`vm.args`

`sys.config`

`REPLACE_OS_VARS`

*.appup Generator



1.2.3

1.2.3

c7ae2

1523874000-c7ae2

```
git rev-list \
  -1 HEAD \
  --format=%ct-%H -- \
  apps/sub_umbrella
| grep -v ^commit
```

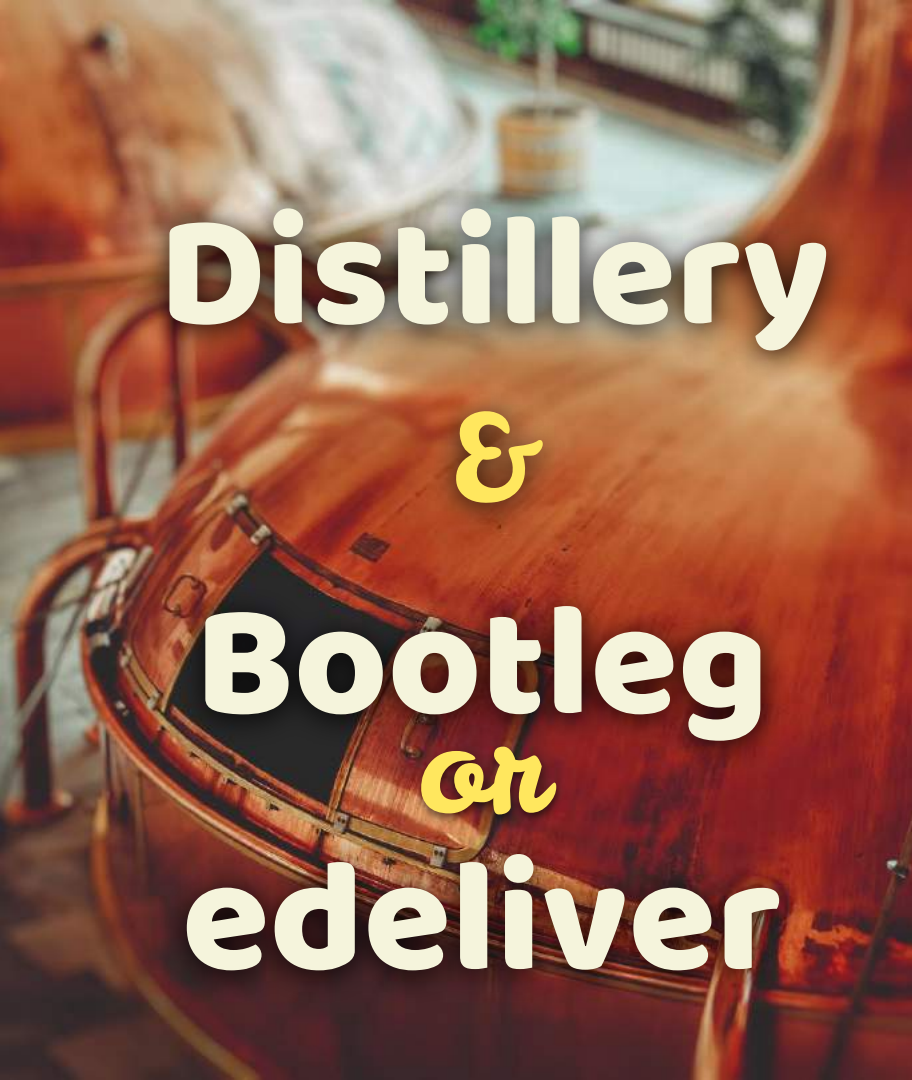



Automate





Gigalixir

A large, polished wooden distillation still, likely a pot still, is the central focus of the image. It has a curved, barrel-like shape with metal bands and a small spout at the bottom. The background is slightly blurred, showing other distillery equipment and a clean, industrial environment.

Distillery

&

Bootleg

or

edeliver

A person wearing a white racing helmet and a dark blue shirt is kneeling on the ground, working on the front of a Formula 1 car. The car is blue and yellow, with 'P ZERO' visible on the tires. Another person in a dark blue shirt is standing in the background. The scene is set in a garage or pit lane.

Gigalixir

beamup[^]



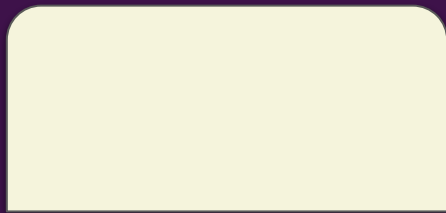
store



you



prod



store



you



prod

\$ curl https://get.beamup.io/install | /bin/sh



store



you



prod



\$ beamup store

BEAMUP_STORE=https://store

\$

beamup build



BEAMUP_NODE_RELEASE=hey

BEAMUP_STORE=https://store

\$

beamup node





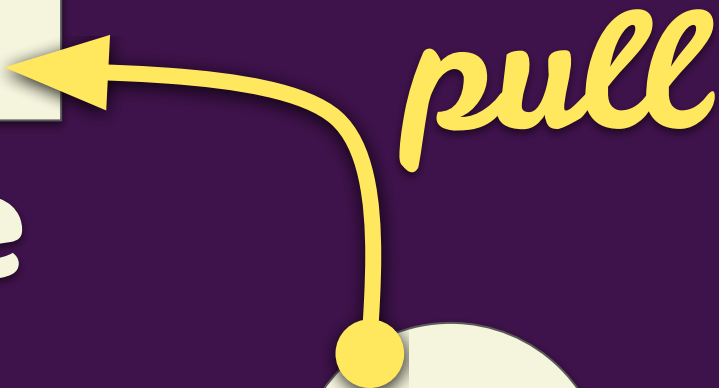
you

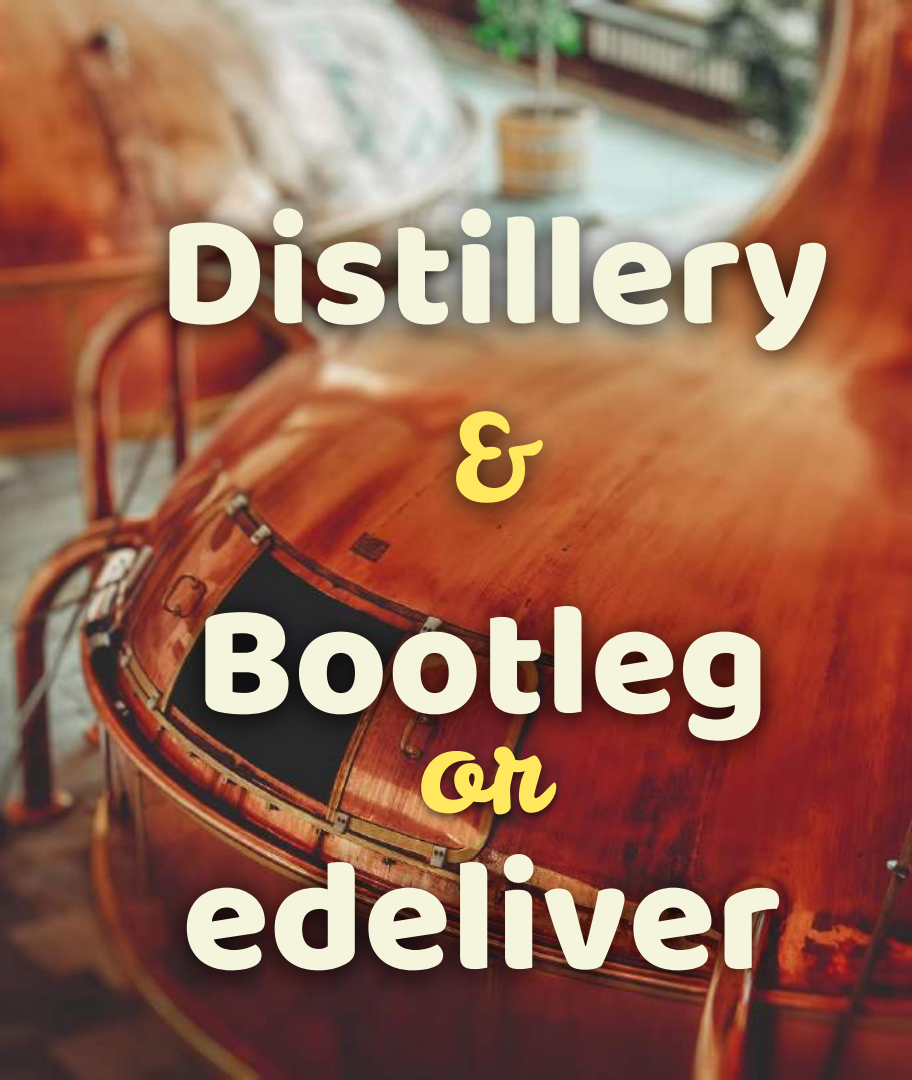


store



prod





Distillery

&

Bootleg

or

edeliver



Gigalixir

Thank
you

The background of the image is a night scene of a snowy forest. The ground is covered in a layer of snow, and several tall, dark evergreen trees are scattered across the landscape. The sky is dark, but a vibrant aurora borealis is visible, with colors ranging from deep purple and blue to bright green and yellow. The word "Questions" is written in a large, yellow, cursive font across the center of the image.

Questions

Thank
you

Photo Credits

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Clem Onojeghuo**

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